

Tutorial 3

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Question 1

Let $x \in R^n$. Its p-norm is defined as

$$\|x\|_p = [\sum_{i=1}^n |x_i|^p]^{1/p}$$

Prove it is a norm when $p \geq 1$.

Hint: The most difficult part is to prove $\|x\|_p + \|y\|_p \geq \|x + y\|_p$.

- When $\|x\|_p, \|y\|_p > 0$. Let $x^1 = \frac{x}{\|x\|_p}$, $y^1 = \frac{y}{\|y\|_p}$. $\lambda = \frac{\|x\|_p}{\|x\|_p + \|y\|_p}$. Then $\|x^1\|_p = \|y^1\|_p = 1$, $0 < \lambda < 1$.

Then $\|x\|_p + \|y\|_p \geq \|x + y\|_p$ is equivalent to

$$\|\lambda x^1 + (1 - \lambda)y^1\|_p \leq 1$$

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$$\|\lambda x^1 + (1 - \lambda)y^1\|_p^p \leq 1$$

Noting that $\|x\|_p^p = |x_1|^p + \dots + |x_n|^p$ is convex.

$$\|\lambda x^1 + (1 - \lambda)y^1\|_p^p \leq \lambda \|x^1\|_p^p + (1 - \lambda) \|y^1\|_p^p = 1$$

- when one of $\|x\|_p$ and $\|y\|_p$ is zero. it is trivial.

Question 2

Let $p(x)$ and $q(x)$ are two probability distributions of a discrete random variable x on X . That is, both $p(x)$ and $q(x)$ sum up to 1, and $p(x) > 0$ and $q(x) > 0$. The Kullback-Leibler divergence is defined as

$$DKL(p(x), q(x)) = \sum_{x \in X} p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq 0$$

Prove $DKL(p(x), q(x)) \geq 0$.

Hint: treat $p(x) = (p_1, \dots, p_n)$ and $q(x) = (q_1, \dots, q_n)$

$\ln(x)$ is concave, so

$$\ln\left(\sum p_i x_i\right) \geq \sum p_i \ln(x_i)$$

If $x_i = \frac{q_i}{p_i}$, then $\sum p_i x_i = 1$.